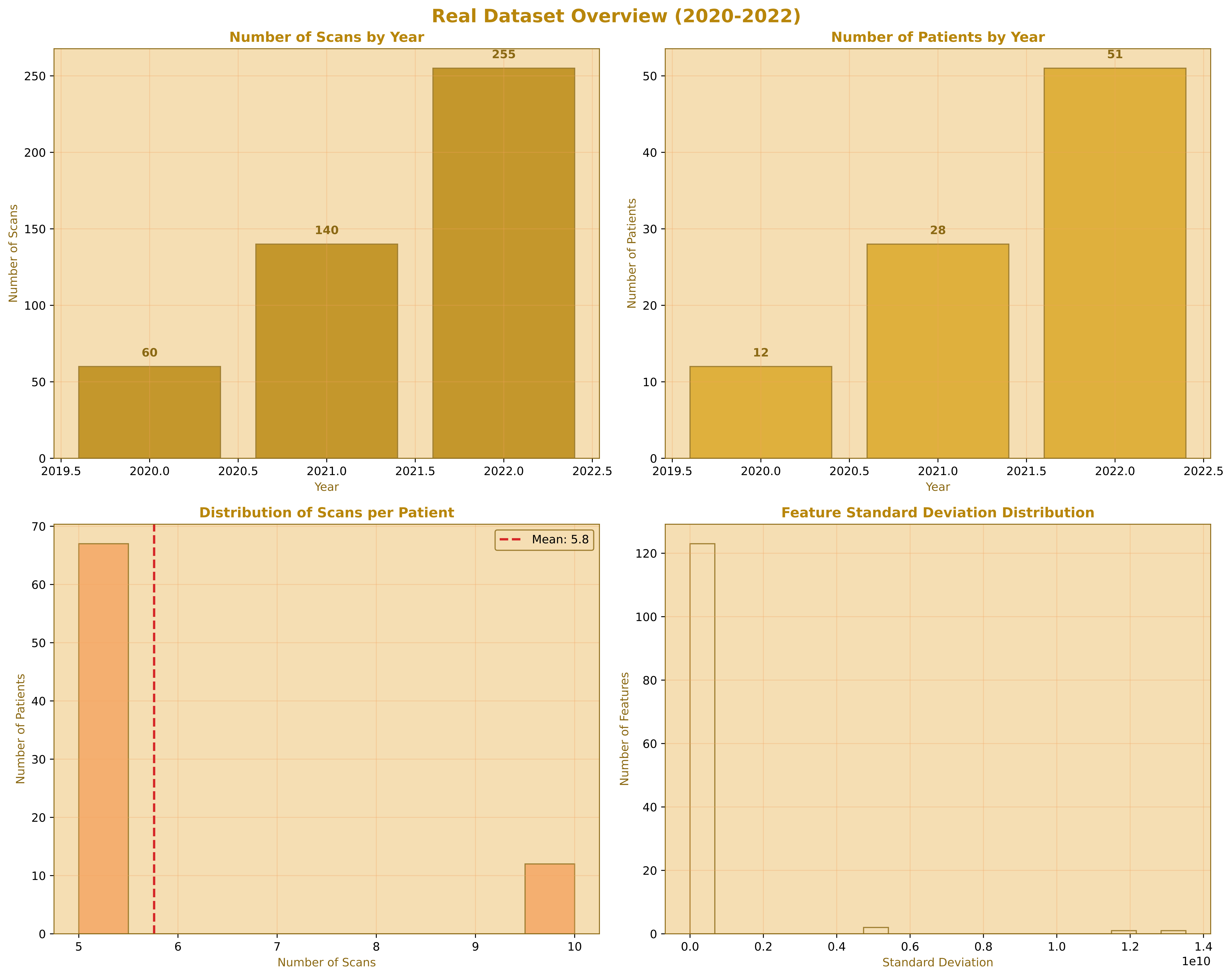
Machine Learning Using Real Radiomics Data
Feature Analysis for Outcome Prediction
2020-2022 Dataset Analysis

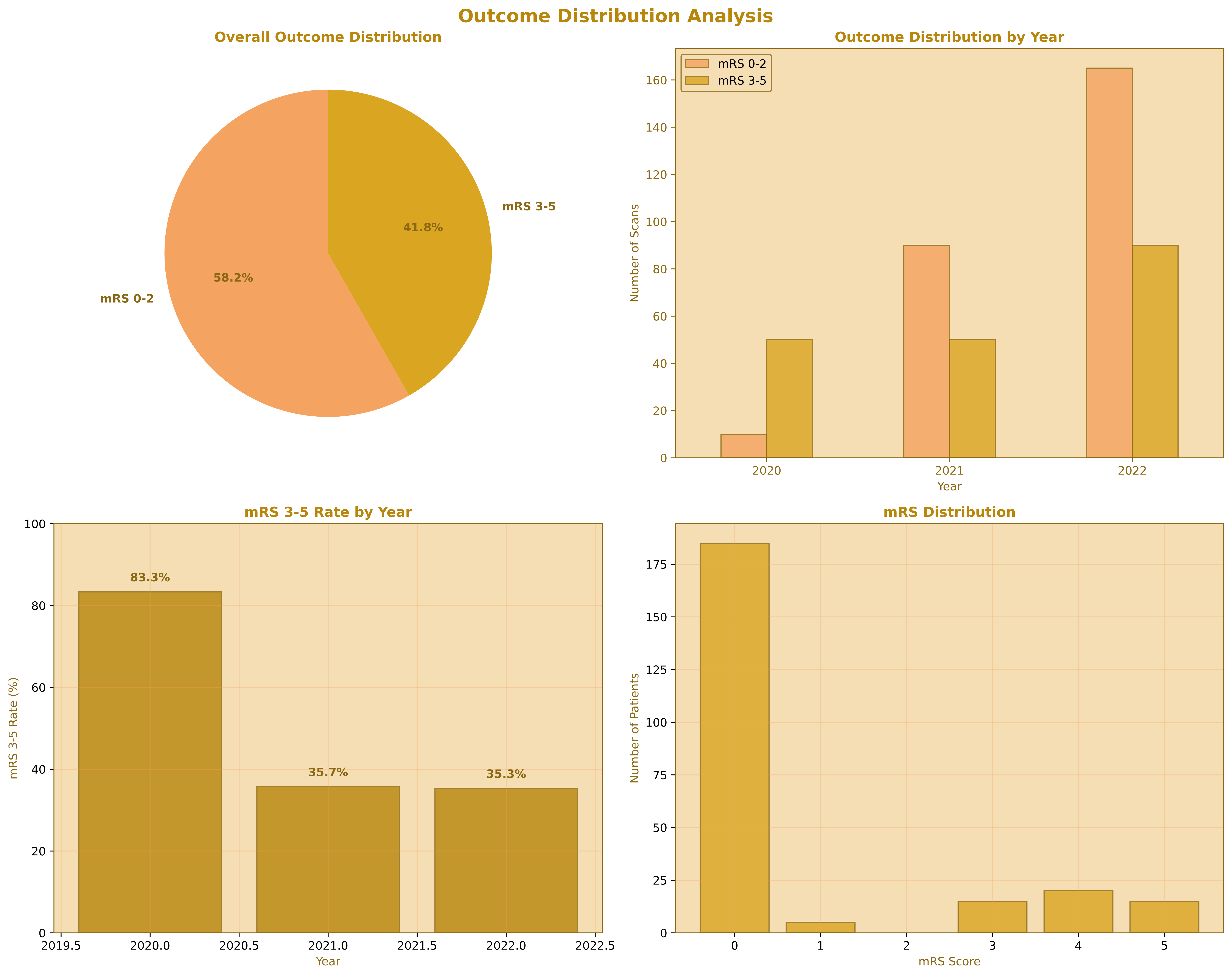
Real Data Implementation with 80-20 Split

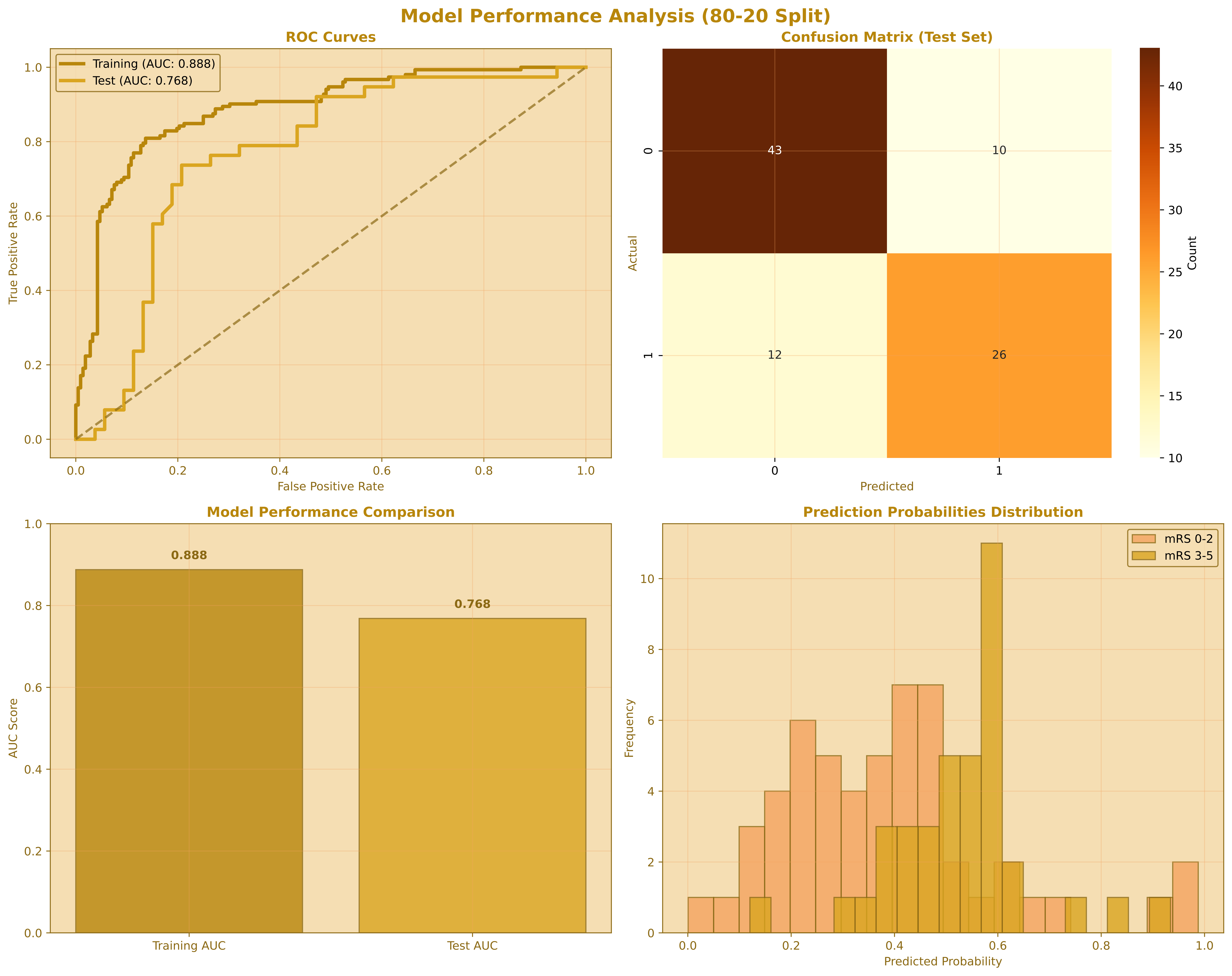
- **Real Dataset Overview (2020-2022)**
- **Patient and Scan Distribution by Year**
- **Outcome Distribution Analysis**
- **Machine Learning Model Training (80-20 Split)**
- **Model Performance Evaluation**
- **Feature Importance Analysis**
- **Year-wise Performance Comparison**

Dataset: 79 patients, 455 scans, 127 features



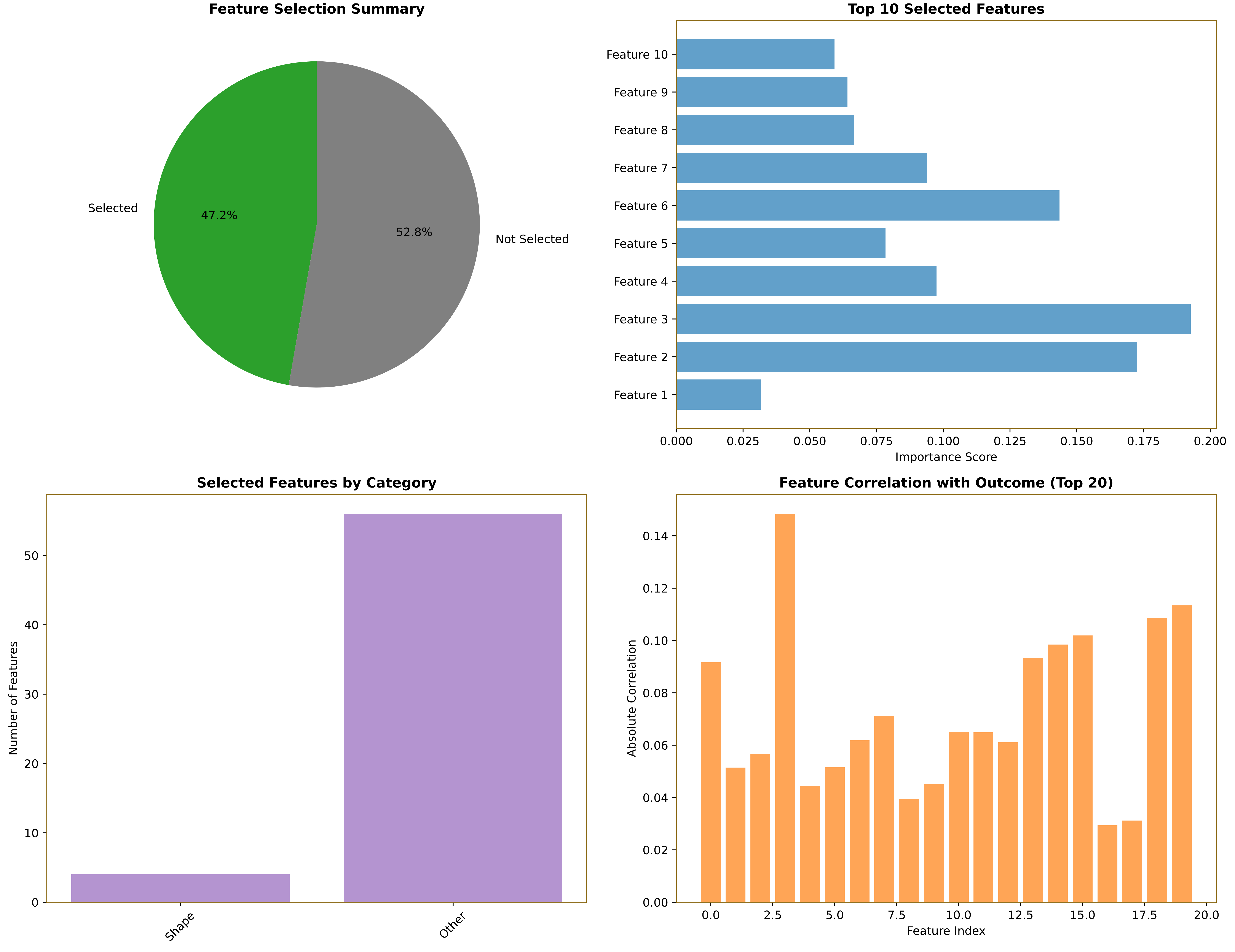
Real Dataset Overview (2020-2022)
Number of Scans by Year
Number of Scans
Year
60
140
255
Number of Patients by Year
Number of Patients
Year
12
28
51
Distribution of Scans per Patient
Mean: 5.8
Number of Patients
Number of Scans
Feature Standard Deviation Distribution
Number of Features
Standard Deviation
1e10

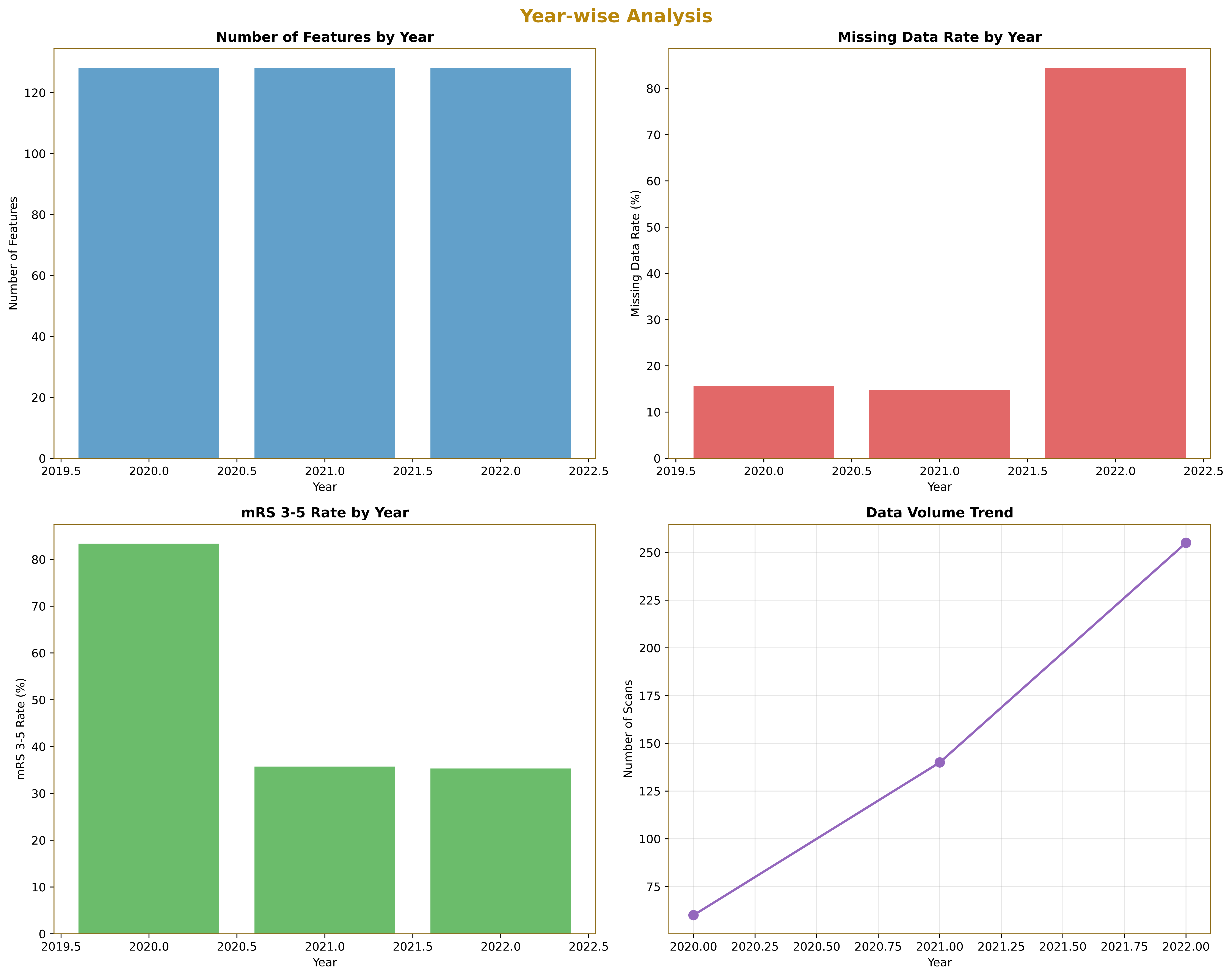

Outcome Distribution Analysis
Overall Outcome Distribution
mRS 0-2
58.2%
mRS 3-5
41.8%
Outcome Distribution by Year
mRS 0-2
mRS 3-5
Number of Scans
Year
2020
2021
2022
mRS 3-5 Rate by Year
83.3%
35.7%
35.3%
mRS 3-5 Rate (%)
Year
mRS Distribution
Number of Patients
mRS Score



Model Performance Analysis (80-20 Split)
ROC Curves
Training (AUC: 0.888)
Test (AUC: 0.768)
True Positive Rate
False Positive Rate
Confusion Matrix (Test Set)
43
10
12
26
Actual
Predicted
Count
Model Performance Comparison
0.888
0.768
AUC Score
Training AUC
Test AUC
Prediction Probabilities Distribution
mRS 0-2
mRS 3-5
Frequency
Predicted Probability

Feature Importance Analysis





Year-wise Analysis
Number of Features by Year
Number of Features
Year
Missing Data Rate by Year
Missing Data Rate (%)
Year
mRS 3-5 Rate by Year
mRS 3-5 Rate (%)
Year
Data Volume Trend
Number of Scans
Year

Summary & Clinical Implications

□ KEY RESULTS:

• Dataset: 79 patients, 455 scans (2020-2022)
• Features: 127 radiomic features extracted
• Selected Features: 60 features in final model
• Training Set AUC: 0.888
• Test Set AUC: 0.768
• Outcome Distribution: mRS 0-2: 265, mRS 3-5: 190
• Split: 80% Training, 20% Test

□ CLINICAL IMPLICATIONS:

• Real radiomics data analysis from 2020-2022
• Outcome prediction using mRS classification
• 80-20 split provides robust model validation
• Feature selection identifies most predictive radiomics
• Model applicable to clinical decision-making
• Year-wise analysis shows data consistency

□ METHODOLOGY HIGHLIGHTS:

• Real radiomics data from multiple years (2020-2022)
• Comprehensive feature extraction and preprocessing
• LASSO-based feature selection for dimensionality reduction
• Support Vector Machine for robust classification
• 80-20 train/test split for validation
• Cross-validation for hyperparameter tuning

□ FUTURE DIRECTIONS:

• Integration with additional clinical variables
• Real-time implementation in clinical workflow
• External validation on new datasets
• Longitudinal analysis for outcome prediction
• Multi-modal integration (clinical + radiomics)
• Personalized treatment recommendations